

ELECTRICAL SYSTEMS INSPECTION REPORT

Quarterly Inspection - Q3 2026 | Unit 4 - Crude Distillation Unit (CDU)

Inspection Date:	September 12, 2026
Report Number:	MRPL/ELE/2026/Q3/047
Inspected By:	Er. R. K. Sharma, Chief Electrical Engineer
Area / Section:	Unit 4 - CDU Electrical Substation & Field Equipment
Classification:	CONFIDENTIAL - For Internal Use Only

1. EXECUTIVE SUMMARY

A comprehensive quarterly inspection of the Unit 4 Crude Distillation Unit electrical systems was conducted on September 12, 2026. The inspection covered the main switchgear room, MCC panels, field instruments, earthing systems, and emergency lighting. A total of 11 discrepancies were identified, of which 3 are classified HIGH severity and require immediate corrective action. Production continuity risk is assessed as MODERATE-HIGH until the main switchgear issue is resolved.

2. KEY FINDINGS

HIGH	F-01 Main 11kV switchgear panel (Panel SG-4A) shows thermal discoloration and carbon tracking on Bus Bar Section B. Internal temperature measured at 78degC against rated 55degC. Risk of arc flash or busbar failure under load.
HIGH	F-02 Motor Control Centre MCC-4B: Contactor for Pump P-401A shows severe pitting and contact erosion. Estimated remaining life < 2 weeks under current operational load. Pump is critical for crude feed.
HIGH	F-03 Earthing resistance at transformer TR-4C measures 14.8 ohms (acceptable limit: <= 5 ohms). Inadequate earthing poses serious personnel safety risk and equipment damage potential during fault conditions.
MEDIUM	F-04 Cable trays in sections E-4 through E-7 showing corrosion-induced structural weakening. 6 cable tray supports observed with >40% cross-section loss due to H2S-induced corrosion.
MEDIUM	F-05 Emergency lighting system: 4 of 22 luminaires in the CDU pump area found non-functional. Does not meet IS:3646 requirements for emergency egress illumination. Replacement bulbs requisitioned.
MEDIUM	F-06 UPS system for control room (UPS-CR-01) shows battery bank health at 62% (threshold: 80%). Auto-changeover test failed on 2nd attempt. Battery replacement recommended before October 2026.
MEDIUM	F-07 Conduit sealing fittings EF-4-101 through EF-4-108 (Hazardous Zone 1) show cracking in sealing compound. Fire and explosion propagation risk in the event of gas release.
LOW	F-08 Cable identification tags missing or faded on 23 cables in Junction Box JB-4-12. Increases maintenance time and misidentification risk during emergency isolation.
LOW	F-09 Motor terminal box covers for M-403B and M-407A show loose fasteners. Ingress protection rating compromised - IP65 no longer assured.
LOW	F-10 Control panel CP-4-02 door seal shows 15cm gap due to hinge damage. Dust and moisture ingress possible; may affect relay calibration long-term.

LOW

F-11 Annunciation panel AP-4-01: 3 alarm points found latched in unacknowledged state for >72 hours. Operational response procedure not followed per SOP-ELE-012.

3. RISK ASSESSMENT

CRITICAL RISK: Simultaneous failure of SG-4A switchgear and MCC-4B contactor could result in complete loss of power to the CDU pump area, leading to unplanned shutdown of crude distillation and estimated production loss of 450 MT/day (approximately INR 2.7 Cr/day). Arc flash hazard at SG-4A poses serious risk of personnel injury to maintenance staff. HIGH RISK: Earthing failure at TR-4C poses risk of equipment damage and personnel electrocution. MEDIUM RISK: Conduit seal failures in Zone 1 areas could result in gas leak propagation. LOW RISK: Documentation and minor maintenance issues - no immediate production impact.

4. RECOMMENDATIONS

R-01 [IMMEDIATE - Within 24 Hours]

De-energize SG-4A Bus Section B and arrange OEM inspection. Isolate from service and switch load to standby SG-4B. Do not restore until thermal scan confirms no hot spots above 60degC.

R-02 [IMMEDIATE - Within 48 Hours]

Replace contactor assembly in MCC-4B for pump P-401A. Procure spare from central stores (part no. ABB-A75-30-11). Arrange shutdown window with Operations.

R-03 [Within 7 Days]

Commission earthing remediation at TR-4C. Engage MRPL Civil dept. for additional earth pits (minimum 3 additional pits required to achieve <= 5 ohm).

R-04 [Within 14 Days]

Replace all 4 non-functional emergency luminaires. Conduct IS:3646 compliant lux-level test and submit compliance certificate to Safety Dept.

R-05 [Within 30 Days]

Replace UPS battery bank UPS-CR-01. Schedule maintenance window with IT/Instrumentation. Conduct load test after replacement to verify auto-changeover within 20ms.

R-06 [Within 30 Days]

Re-seal all 8 conduit sealing fittings in Zone 1 (EF-4-101 to EF-4-108). Use EX-rated compound grade M60. Verify against zone classification drawing.

5. SIGN-OFF & APPROVALS

Inspected By	Er. R. K. Sharma	Chief Electrical Engineer
Reviewed By	Er. P. N. Bhat	DGM - Electrical Maintenance
Approved By	Mr. S. K. Menon	VP - Operations & Maintenance

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